

TECHNICAL ADDENDUM

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SUBJECT: **Mathematical Justification: Algorithm Selection for Time-To-Event Attrition Modeling**

1. Executive Summary

This document outlines the rigorous mathematical rationale for selecting XGBoost formulated for Survival Analysis as the primary predictive engine for the HR Attrition model. The justification is derived from three core mathematical constraints: the requirement for a censored likelihood function, the violation of classical proportionality and parametric assumptions by the empirical data distribution, and the superior bias-variance tradeoff mechanics of functional gradient descent over independent bootstrap aggregation.

2. Formulation of the Predictive Problem

Let the true unobserved survival time for employee i be a random variable T_i , and the censoring time be C_i . The observed time is $Y_i = \min(T_i, C_i)$. The censoring indicator is defined as $\delta_i = I(T_i \leq C_i)$, where $\delta_i = 1$ if attrition is observed, and 0 if censored (active employee).

The feature space is defined by our 32 encoded attributes, represented as a vector $X_i \in \mathbf{R}^{32}$. The business objective requires estimating the conditional survival function at specific temporal intervals:

$$S(t | X_i) = P(T_i > t | X_i) \quad \text{for } t \in \{1, 3, 6, 12\}$$

3. Invalidation of Standard Regression Models

Standard classification models (e.g., Logistic Regression) estimate $P(\delta_i = 1 | X_i)$ by maximizing the Bernoulli log-likelihood:

$$L_{reg}(\theta) = \sum_{i=1}^n [\delta_i \log(p_i) + (1 - \delta_i) \log(1 - p_i)]$$

where $p_i = \sigma(\theta^T X_i)$.

Mathematical Failure: This formulation is inconsistent for right-censored data. For censored observations ($\delta_i = 0$), the likelihood term reduces to $\log(1 - p_i)$. This mathematically assumes that the event *did not occur* at time Y_i , but completely discards the information that the employee *survived up to* time Y_i . It equates an employee censored at $t=1$ month with an employee censored at $t=120$ months. To

generate accurate time-horizon probabilities, we must maximize a likelihood that integrates the survival function $S(Y_i | X_i)$ for censored observations, which standard regression mathematically cannot support.

4. Limitations of Classical Survival Models

While Survival Analysis resolves the censoring issue via the likelihood $L_{surv} = \sum [\delta_i \log h(Y_i | X_i) + \log S(Y_i | X_i)]$, classical parameterizations impose rigid constraints on the hazard function $h(t | X)$.

A. Cox Proportional Hazards (Cox PH)

The Cox model specifies: $h(t | X) = h_0(t) \exp(\beta^T X)$.

- **Mathematical Failure:** The model is bound by the Proportional Hazards (PH) assumption. For any two feature vectors X_a, X_b , the log-hazard ratio must be time-invariant:

$$\partial/\partial t \log [h(t | X_a) / h(t | X_b)] = \partial/\partial t [\beta^T (X_a - X_b)] = 0 \quad \forall t$$

In the HR feature space, this assumption is empirically violated. The partial derivative of the hazard with respect to a feature (e.g., OverTime) is not constant over time; the risk of burnout accelerates non-linearly relative to tenure. When the PH assumption is violated, the partial likelihood estimator β becomes biased, rendering the projected $S(t)$ estimates at 6 and 12 months mathematically inconsistent.

B. Parametric Models (Weibull)

The Weibull model assumes a baseline hazard of the form: $h_0(t) = (\rho/\lambda) (t/\lambda)^{\rho-1}$.

- **Mathematical Failure:** The monotonicity of the hazard function is entirely determined by the shape parameter ρ . If $\rho > 1$, $h(t)$ is strictly monotonically increasing; if $\rho < 1$, strictly decreasing. HR attrition data frequently exhibits non-monotonic baseline hazards (e.g., local maxima corresponding to stock vesting cliffs or annual review cycles). Because the MLE cannot map a Weibull distribution to a function with local maxima/minima, the asymptotic bias of the estimator increases significantly, failing to capture critical inflection points in the 12-month timeline.

5. Superiority of XGBoost over Tree Ensembles

To resolve the bias of Cox/Weibull without introducing the censoring failures of regression, we utilize non-parametric tree-based methods.

A. Failure of Single Decision Trees

A single survival tree partitions the feature space $\mathbf{R}^{32}$ into M orthogonal regions $R_1, \dots, R_M$. To capture high-order interactions (e.g., $X_{overtime} \cap X_{income} \cap X_{satisfaction}$), the tree must generate highly complex, non-linear partitions. In a 32-dimensional space, this requires unbounded tree depth $d \rightarrow \infty$, leading to high variance: $Var(S(t|X)) \rightarrow \infty$ (overfitting).

B. Sub-optimality of Random Survival Forests (RSF)

RSF applies bootstrap aggregation (bagging) to reduce variance: $S_{RF}(t|X) = (1/B) \sum_{b=1}^B S_b(t|X)$.

- **Mathematical Failure:** Bagging is highly effective at reducing variance ($Var \approx \rho \sigma^2 / B$, where ρ is tree correlation), but it fundamentally cannot reduce model bias. Because each tree b is fitted independently, Tree b has no mathematical mechanism to correct the systematic errors of Tree $b-1$. To approximate complex interactions, RSF must grow deep trees, which inherently increases tree correlation $\rho \rightarrow 1$, thereby neutralizing the variance reduction formula.

C. Mathematical Optimization of XGBoost (Gradient Boosting)

XGBoost formulates the problem as functional optimization in an additive Hilbert space:

$$\hat{f}^*(X) = \underset{f}{\operatorname{argmin}} \sum_{i=1}^n L(y_i, f(X_i)) + \Omega(f)$$

where L is a differentiable convex survival loss (e.g., Cox Partial Likelihood) and $\Omega(f) = \sum_{m=1}^M y T_m + \frac{1}{2} \lambda \sum_{j=1}^T w_{mj}^2$ is the regularization penalty on tree complexity.

At iteration k , the algorithm fits a new tree $h_k(X)$ via Forward Stagewise Additive Modeling:

$$\hat{f}_k(X) = \hat{f}_{k-1}(X) + \eta h_k(X)$$

Crucially, h_k is not fitted to the raw data, but to the negative gradient (pseudo-residuals) of the loss function evaluated at the previous step:

$$r_{ik} = - \left[\partial L(y_i, \hat{f}_{k-1}(X_i)) / \partial \hat{f}_{k-1}(X_i) \right] \quad f = \hat{f}_{k-1}$$

Why XGBoost Mathematically Dominates:

1. **Sequential Bias Reduction:** By calculating the gradient ∇L at step $k-1$, the algorithm identifies the exact direction in the 32-dimensional feature space where the current additive estimate $\hat{f}_{k-1}$ has the highest error. Tree h_k is mathematically constrained to fit this specific residual subspace. This functional gradient descent allows XGBoost to systematically isolate high-order feature interactions (the non-proportional hazards) that RSF cannot see due to independent fitting.
2. **Optimal Bias-Variance Tradeoff:** The regularization term $\Omega(f)$ penalizes the L_2 norm of the leaf weights w and the number of terminal nodes T . This allows XGBoost to grow deep enough to capture complex interactions (reducing bias, overcoming the Cox PH flaw) while mathematically shrinking the leaf weights (controlling variance, overcoming the Single Tree flaw).

6. Conclusion

The selection of XGBoost Survival is not arbitrary; it is the only evaluated algorithm that satisfies the mathematical prerequisites of the HR predictive problem. It utilizes a censored likelihood function (eliminating the bias of standard regression), operates as a non-parametric model (eliminating the monotonicity and proportionality constraints of Weibull and Cox), and leverages functional gradient descent with L_2 regularization (optimizing the bias-variance tradeoff in a 32-dimensional interaction space, overcoming the limitations of bagged ensembles).